INFORMATICS INSTITUTE OF TECHNOLOGY

CW1 – GROUP PROJECT REPORT

3D Interactive Media Product

THE LOST ISLAND

Group Details

Contributor	Student identifier	Main responsibility
Student 1 – K. K. Hettiarachchi	20241840/ w2182374	Beach starting area; physics-based LEGO brick; minifigure walking button
Student 2 – Ranasinghe Arachchige Sayuni Hasara	20232913 / w2182333	Jungle environment; mouse-controlled LEGO brick; Stand/Walk/Flip animations
Student 3 and Student 4 – K. Anuk Imira	20241663 / w2182300	Bridge and character movement; ancient temple and trigger-system responsibilities; lobby/scene integration
Student 5 – Methuli Abeysooriya	20232602 / W2153564	Treasure room; proximity trigger; slider-controlled chest; lighting and testing controller

Draft for personal review – replace bracketed notes and add your own screenshots before submission.

1. Introduction

The Lost Island is a LEGO-inspired 3D exploration game developed in Unity for Coursework 1. The project places the player on a mysterious tropical island and guides them through a series of themed environments, including a beach, jungle paths, a bridge and exploration route, ancient ruins, and a treasure location.

The main design aim is to create an exploration-focused experience in which the player moves through connected areas and interacts with objects. The project combines environment construction, physics-based object control, character movement, animation, trigger-based interaction, user-interface controls, lighting and scene navigation. Rather than relying on combat, the experience uses environmental details and interactive objects to support exploration and progression.

The project was developed using Unity 6 (6000.0.55f1) with the Universal Render Pipeline (URP). The final project uses separate scenes, including OpeningLobby, MainIsland, Cave and EndingLobby. Individual work was developed and tested in separate scenes or modules before being adapted for the group project.

2. Project Concept and Gameplay

The player begins from a lobby and enters the island environment. The beach provides the starting area, while the jungle and other environmental sections create routes for exploration. The player character is controlled using keyboard movement and a follow camera. LEGO brick interactions demonstrate physics and alternative input methods. The later sections introduce ancient structures and the treasure-room objective.

The treasure chest provides a more focused interaction. The player approaches the chest, enters a proximity trigger and receives an on-screen prompt. Pressing E opens an interface containing a slider. The slider controls the chest lid position, allowing the player to adjust the opening rather than relying only on a one-click animation. The lobby scenes provide navigation into the game and options to replay or return to the menu.

3. Team Organisation and Individual Contributions

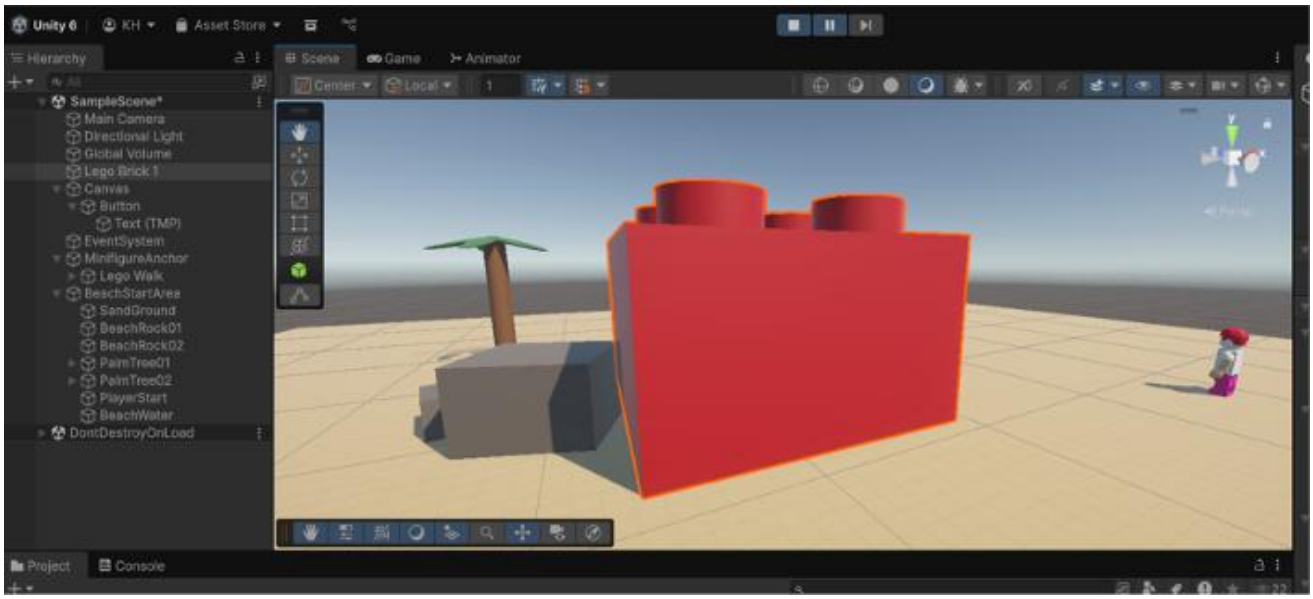
3.1 Student 1 – Beach Starting Area and Physics-Based LEGO Brick

Student 1, K. K. Hettiarachchi (Student ID 20241840), was responsible for the beach starting area and the physics-based LEGO brick task. The beach was constructed using a plane with a sand-coloured material. Rocks were placed beside the brick test route, and palm trees were assembled from simple 3D shapes and positioned around the beach. An empty GameObject named PlayerStart marked the intended starting position. The environment objects were organised under BeachStartArea to make the scene easier to manage and hand over.

For the brick interaction, the LEGO brick model was configured with a Rigidbody and Box Collider. Gravity allowed it to fall onto the ground, while a PhysicsController script applied force in response to WASD input. Testing showed that a speed value of 10 was too low for reliable movement on the sand, so the value was increased to 30. The brick was also tested against a rock to confirm that the collision stopped its movement.

Student 1 also configured a LEGO minifigure walking animation using an Animator Controller with Idle and Take 001 states. A Canvas button called the walking animation through the Animator. A parent

object named MinifigureAnchor was used to keep the model positioned in the camera view, and an Animator transition returned the animation to Idle so it could be triggered again. The beach scene was saved as Student1_Beach and exported as a Unity package with dependencies for group handoff.





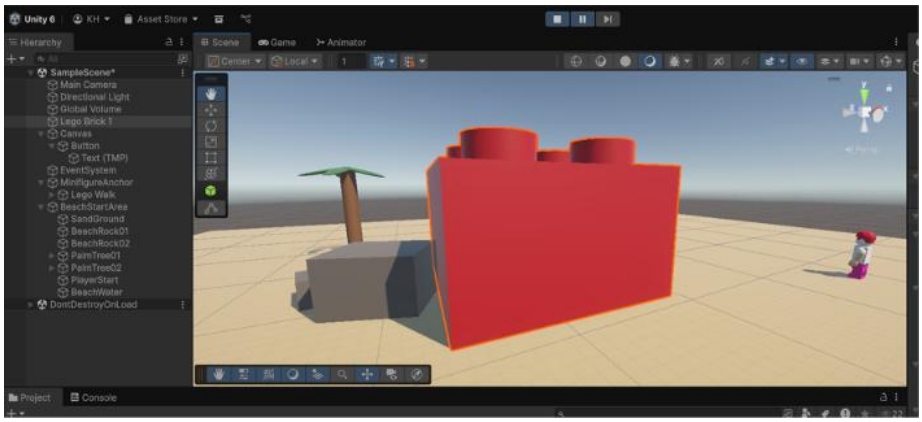
3.2 Student 2 – Jungle Environment and Mouse-Controlled LEGO Brick

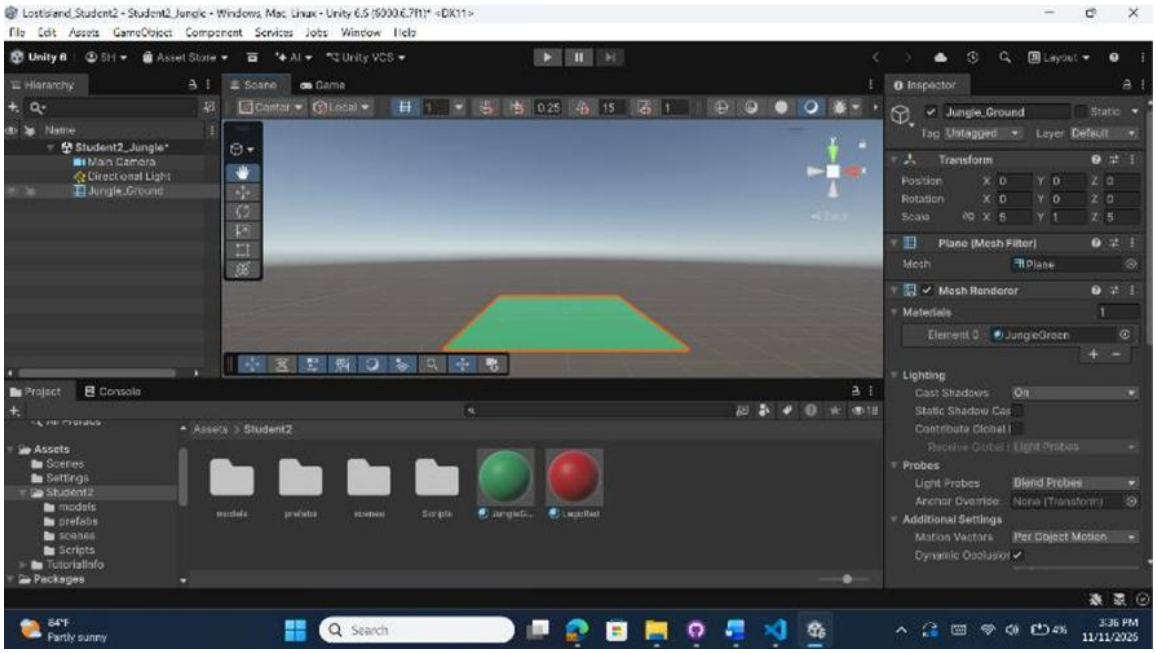
Student 2, Ranasinghe Arachchige Sayuni Hasara (Student ID 20232913; UOW number w2182333), was responsible for the jungle environment, the mouse-input LEGO brick controller and a minifigure animation demonstration.

The jungle prototype was created in a separate scene named Student2_Jungle. A plane formed the ground, a cube-based path provided a route through the area, and green and brown materials helped establish the tropical setting. Palm trees were assembled from basic 3D shapes and arranged around the path. Object placement was adjusted to make the area more organised and suitable for the island theme.

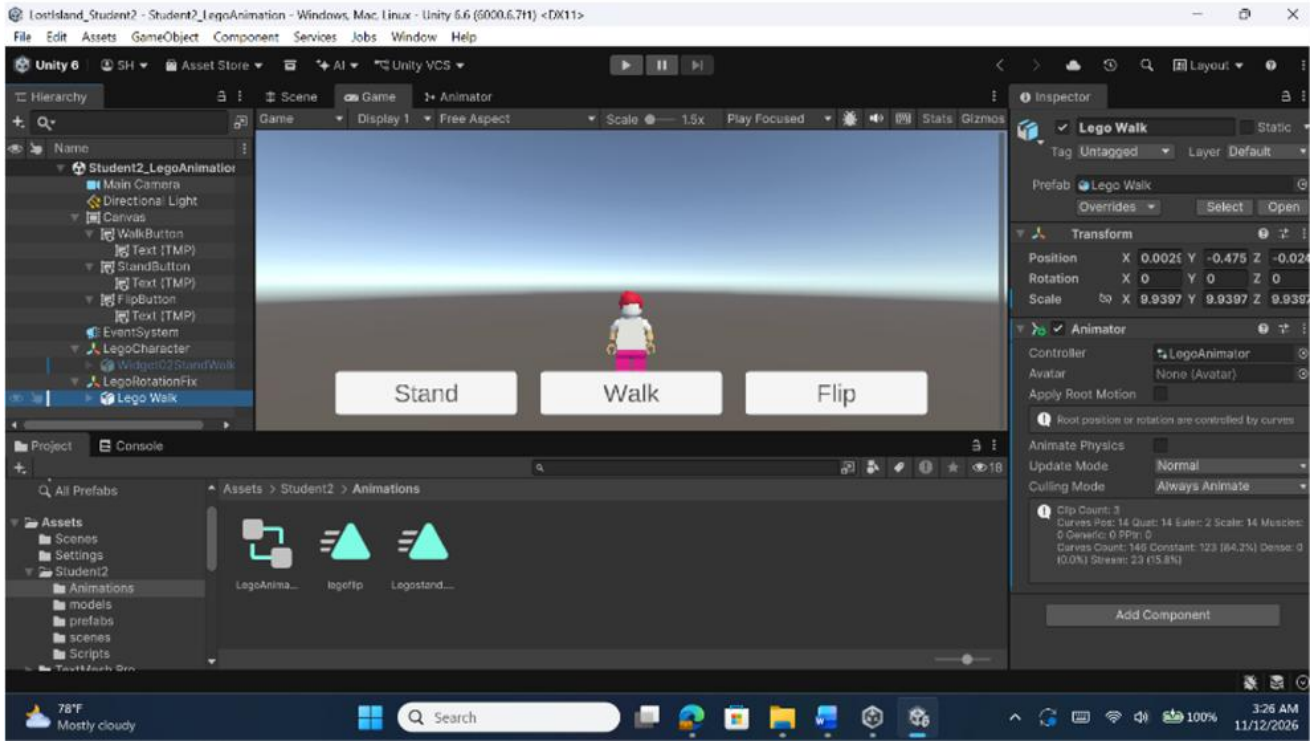
For the mouse-controlled brick, a red LEGO brick was given a Rigidbody and Box Collider. The MouseInput.cs script used mouse dragging in the Game view to control the brick. The feature was tested separately to confirm that the brick responded to mouse input.

Student 2 also created a separate LEGO animation scene using LegoWalk.FBX and an Animator Controller named LegoAnimator. The Stand, Walk and Flip animation states were connected to three UI buttons using the Animator Play function. The report records that the jungle, mouse controller and three animation buttons were tested in Unity.






File Edit Assets GameObject Component Services Jobs Window Help
Hierarchy
Scene
Game
Inspector
Student2_Jungle
Main Camera
Directional Light
Jungle_Ground
Jungle_Path
Tree1
Tree_Trunk
leaf
leaf2
leaf3
leaf4
leaf5
leaf6
leaf7
Cameras
Main Camera
Project
Console
Assets > Student2
Assets
Scenes
Settings
Student2
models
prefabs
scenes
Scripts
TutorialInfo
Packages
LegoRed.mat
Search
84°F
Mostly cloudy
4:41 PM
11/11/2026


LostIsland_Student2 - Student2_LegoAnimation - Windows, Mac, Linux - Unity 6.6 (6000.6.7f1) <DX11>
File Edit Assets GameObject Component Services Jobs Window Help
Hierarchy
Scene
Game
Animator
Inspector
Student2_LegoAnimatio
Main Camera
Directional Light
Canvas
WalkButton
Text (TMP)
StandButton
Text (TMP)
FlipButton
Text (TMP)
EventSystem
LegoCharacter
LegoRotationFix
Lego Walk
Stand
Walk
Flip
Lego Walk
Tag Untagged
Layer Default
Prefab Lego Walk
Transform
Position
Rotation
Scale
Animator
Controller LegoAnimator
Avatar
Apply Root Motion
Animate Physics
Update Mode Normal
Culling Mode Always Animate
Add Component
Project
Console
Assets > Student2 > Animations
Assets
Scenes
Settings
Student2
Animations
models
prefabs
scenes
Scripts
Search
78°F
Mostly cloudy
3:26 AM
11/12/2026

3.3 Student 3 and Student 4 – Bridge, Character Movement, Ancient Temple and Trigger-Based Progression

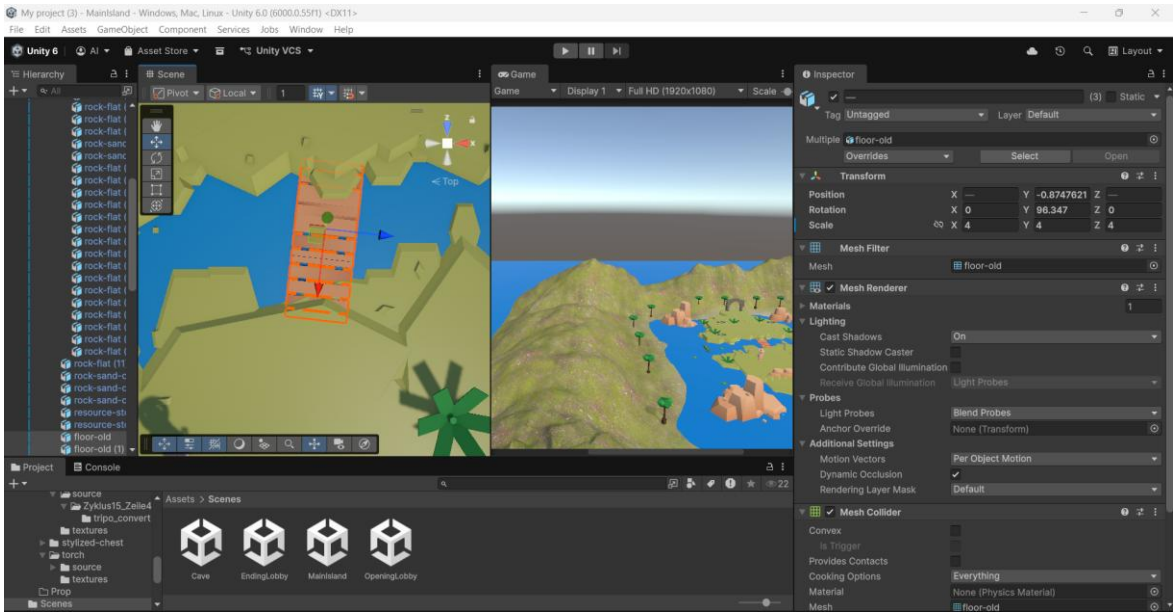
K. Anuk Imira (Student ID 20241663; UOW number w2182300) was assigned Student 3 and Student 4 responsibilities. Student 3 focused on the bridge and character movement, while Student 4 focused on the ancient temple environment, brick mechanisms or obstacles, and trigger-system integration.

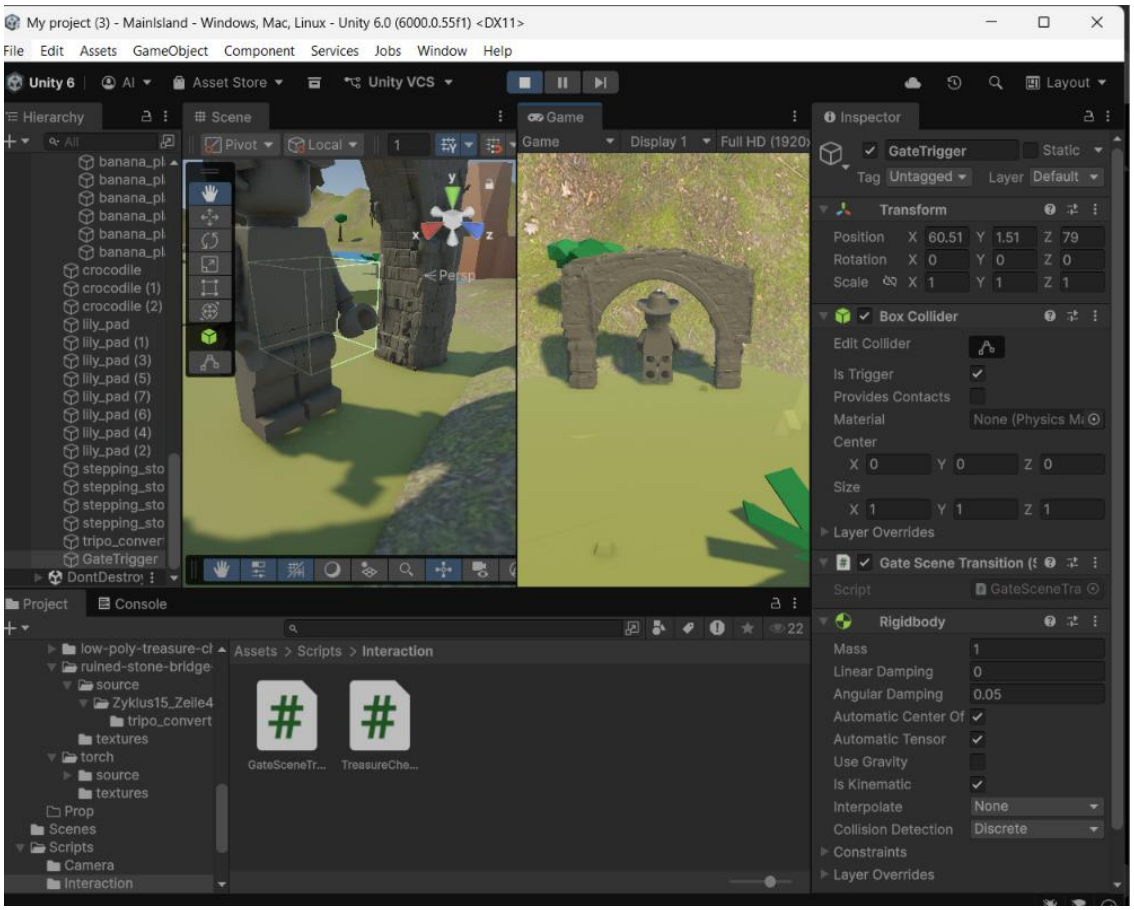
For the integrated project, the player object used Unity's CharacterController component with a PlayerMovement script. Keyboard input controlled movement and gravity was applied so that the character remained grounded. Movement was tested in the Game view. The terrain used its Terrain Collider, and imported structural or land meshes were checked for suitable collision. Mesh Collider components were used where appropriate to prevent the player from passing through solid environment pieces.

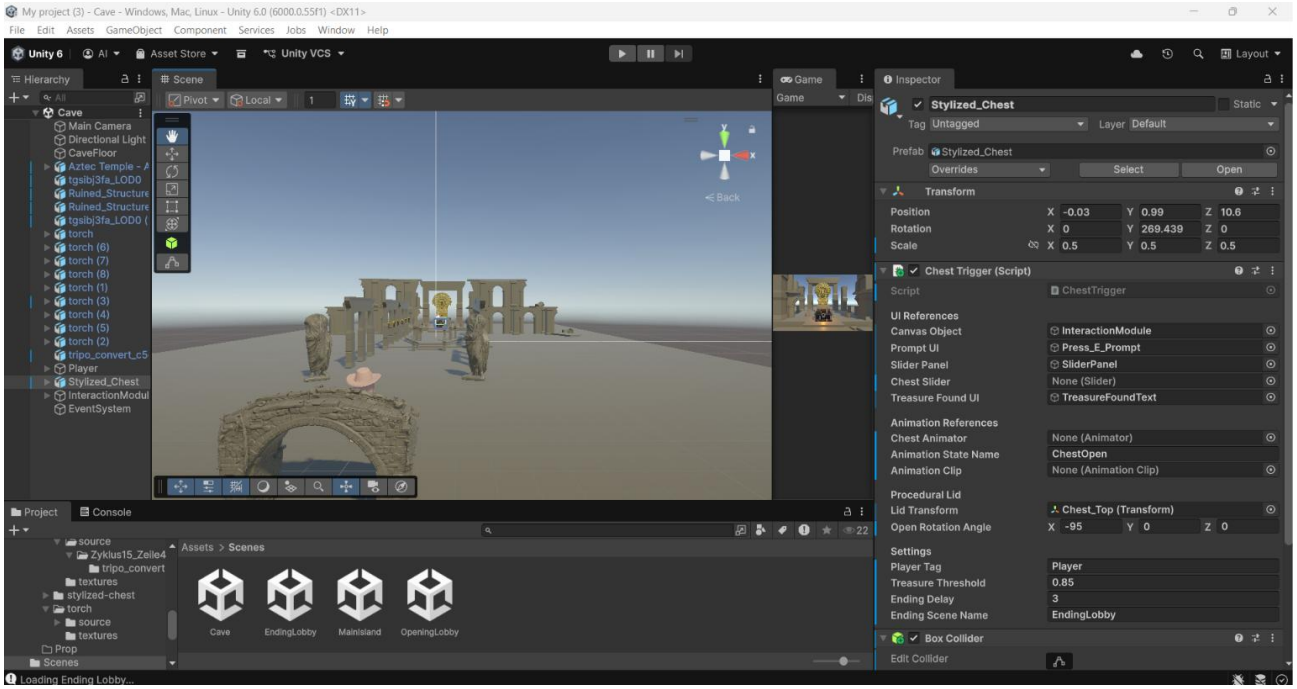
A CameraFollow script was attached to the Main Camera and configured with the Player as its target. The follow offset and smoothing were adjusted to keep the character visible while exploring. The route design also required attention to slopes and transitions: surfaces that were too steep for the CharacterController needed a gentler route or ramp so that the player could traverse them.

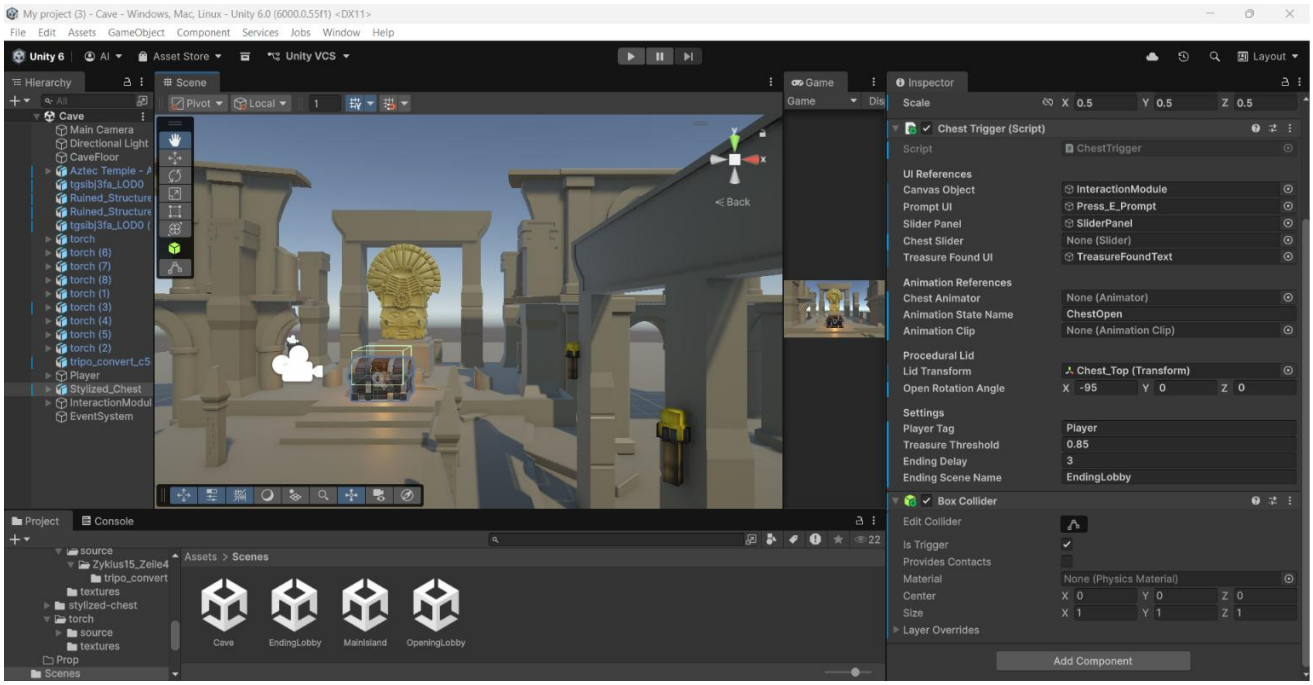
The Student 4 responsibility concerned the ancient temple and its interaction or trigger-based progression. In the combined project, the ancient environment and connected locations were arranged as part of the exploration route. Trigger volumes and collider configuration were considered as part of guiding the player and supporting object interaction. The final report material also describes the chest proximity trigger as an integrated group feature; the dedicated treasure chest system is attributed separately to Student 5 in this report.

Scene navigation was also integrated. The OpeningLobby New Game button loads MainIsland. In EndingLobby, Play Again reloads MainIsland and Main Menu returns to OpeningLobby. These button events use the OpeningLobbyButtons script and Unity's SceneManager.LoadScene method. The scenes were added to Unity 6 Build Profiles so they could be loaded at runtime.






My project (3) - Cave - Windows, Mac, Linux - Unity 6.0 (6000.0.55f1) <DX11>
File Edit Assets GameObject Component Services Jobs Window Help
Unity 6
Asset Store
Unity VCS
Hierarchy
Scene
Game
Inspector
Cave
Main Camera
Directional Light
CaveFloor
Stylized_Chest
InteractionModul
EventSystem
Pivot
Local
Back
Stylized_Chest
Tag Untagged
Layer Default
Prefab Stylized_Chest
Overrides
Select
Open
Transform
Position X -0.03 Y 0.99 Z 10.6
Rotation X 0 Y 269.439 Z 0
Scale X 0.5 Y 0.5 Z 0.5
Chest Trigger (Script)
Script ChestTrigger
UI References
Canvas Object InteractionModule
Prompt UI Press_E_Prompt
Slider Panel SliderPanel
Chest Slider None (Slider)
Treasure Found UI TreasureFoundText
Animation References
Chest Animator None (Animator)
Animation State Name ChestOpen
Animation Clip None (Animation Clip)
Procedural Lid
Lid Transform Chest_Top (Transform)
Open Rotation Angle X -95 Y 0 Z 0
Settings
Player Tag Player
Treasure Threshold 0.85
Ending Delay 3
Ending Scene Name EndingLobby
Box Collider
Edit Collider
Project
Console
Assets > Scenes
Cave
EndingLobby
MainIsland
OpeningLobby
Loading Ending Lobby...


My project (3) - Cave - Windows, Mac, Linux - Unity 6.0 (6000.0.55f1) <DX11>
File Edit Assets GameObject Component Services Jobs Window Help
Unity 6
Asset Store
Unity VCS
Hierarchy
Scene
Game
Inspector
Cave
Main Camera
Directional Light
CaveFloor
Stylized_Chest
InteractionModul
EventSystem
Pivot
Local
Back
Scale X 0.5 Y 0.5 Z 0.5
Chest Trigger (Script)
Script ChestTrigger
UI References
Canvas Object InteractionModule
Prompt UI Press_E_Prompt
Slider Panel SliderPanel
Chest Slider None (Slider)
Treasure Found UI TreasureFoundText
Animation References
Chest Animator None (Animator)
Animation State Name ChestOpen
Animation Clip None (Animation Clip)
Procedural Lid
Lid Transform Chest_Top (Transform)
Open Rotation Angle X -95 Y 0 Z 0
Settings
Player Tag Player
Treasure Threshold 0.85
Ending Delay 3
Ending Scene Name EndingLobby
Box Collider
Edit Collider
Is Trigger
Provides Contacts
Material None (Physics Material)
Center X 0 Y 0 Z 0
Size X 1 Y 1 Z 1
Layer Overrides
Add Component
Project
Console
Assets > Scenes
Cave
EndingLobby
MainIsland
OpeningLobby

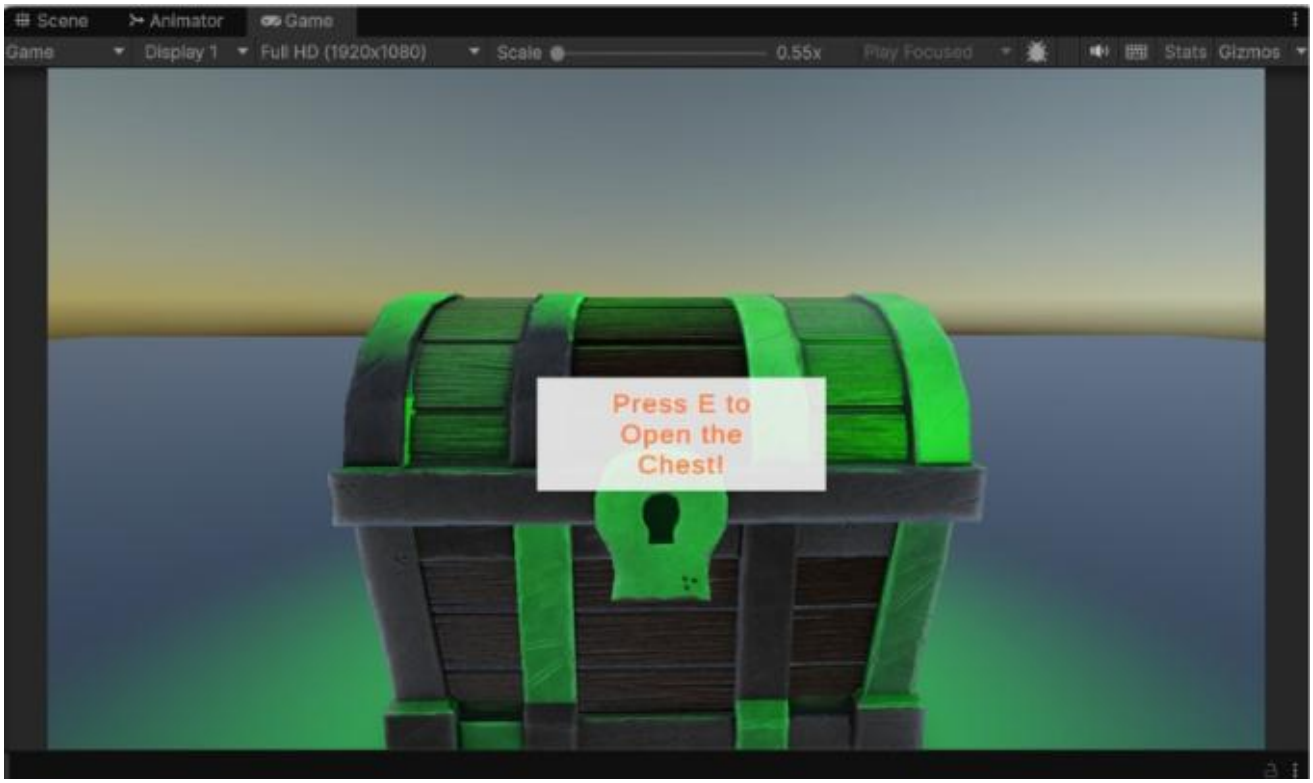
3.4 Student 5 – Treasure Room and Interactive Chest System

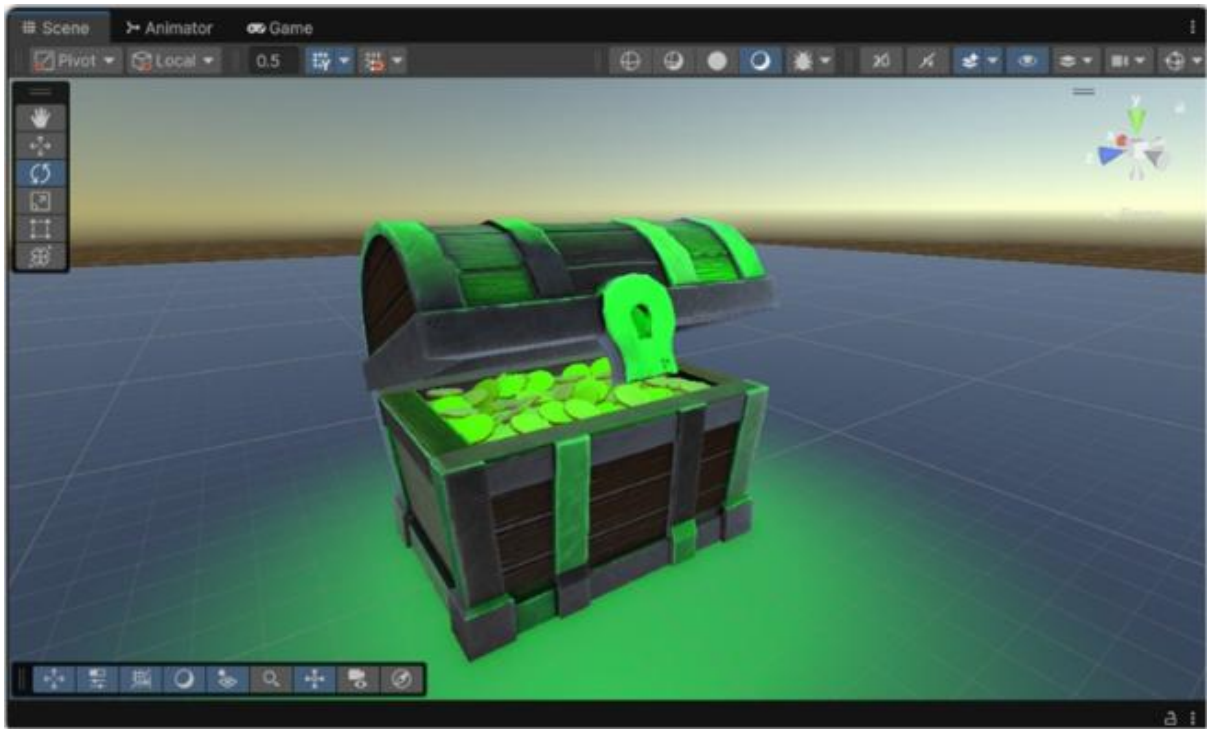
Student 5, Methuli Abeysooriya (IIT ID 20232602; UOW number W2153564), was responsible for the treasure-room scene and the interactive chest system. This contribution included proximity detection, interaction UI, real-time slider control of the chest lid, chest model hierarchy, animation control, a testing character controller and atmospheric lighting.

The chest model was organised into separate components, including Chest_Bottom, Chest_Top, Coins and Coins_Base. Separating the lid from the base allowed the lid to rotate around its hinge while the coins remained inside the chest. The proximity system used ChestTrigger.cs to detect when the player entered the interaction area and display a prompt. ChestTriggerBridge.cs was used to forward trigger events from nested child colliders to the main trigger controller.

The interaction interface used a UI slider with a normalised range from 0 to 1. The slider value controlled the chest opening position. The Animator could be paused and sampled at the slider's normalised time; a Quaternion.Slerp rotation fallback was also described for cases where the Animator controller was unavailable. Cursor handling was adjusted so the player could use the slider without unwanted camera movement.

Student 5 also described a first-person testing controller based on CharacterController, with mouse-look limits and cursor locking. The treasure scene used URP post-processing, including Bloom and Tonemapping, and focused spot lighting to draw attention to the chest and coins. These systems formed the final treasure-room interaction and visual presentation.




Scene
Animator
Game
Pivot
Local
0.5


Scene
Animator
Game
Display 1
Full HD (1920x1080)
Scale
0.55x
Stats
Gizmos

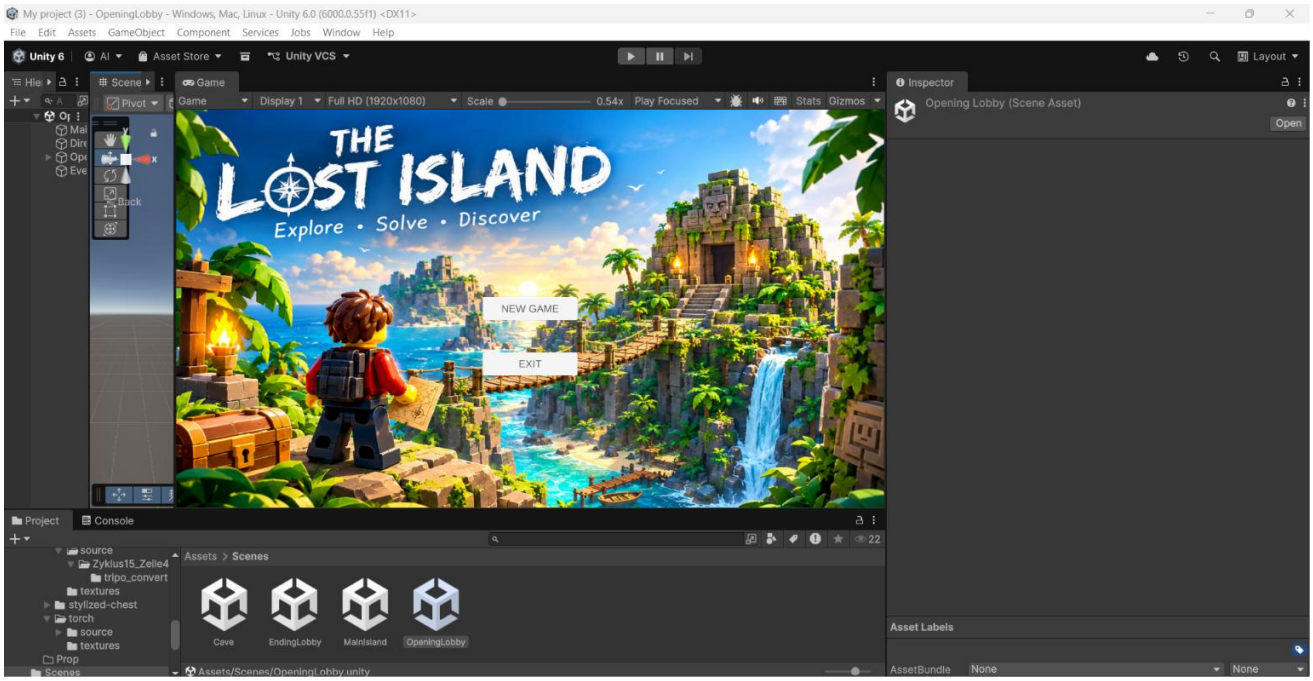
4. Development and Technical Implementation

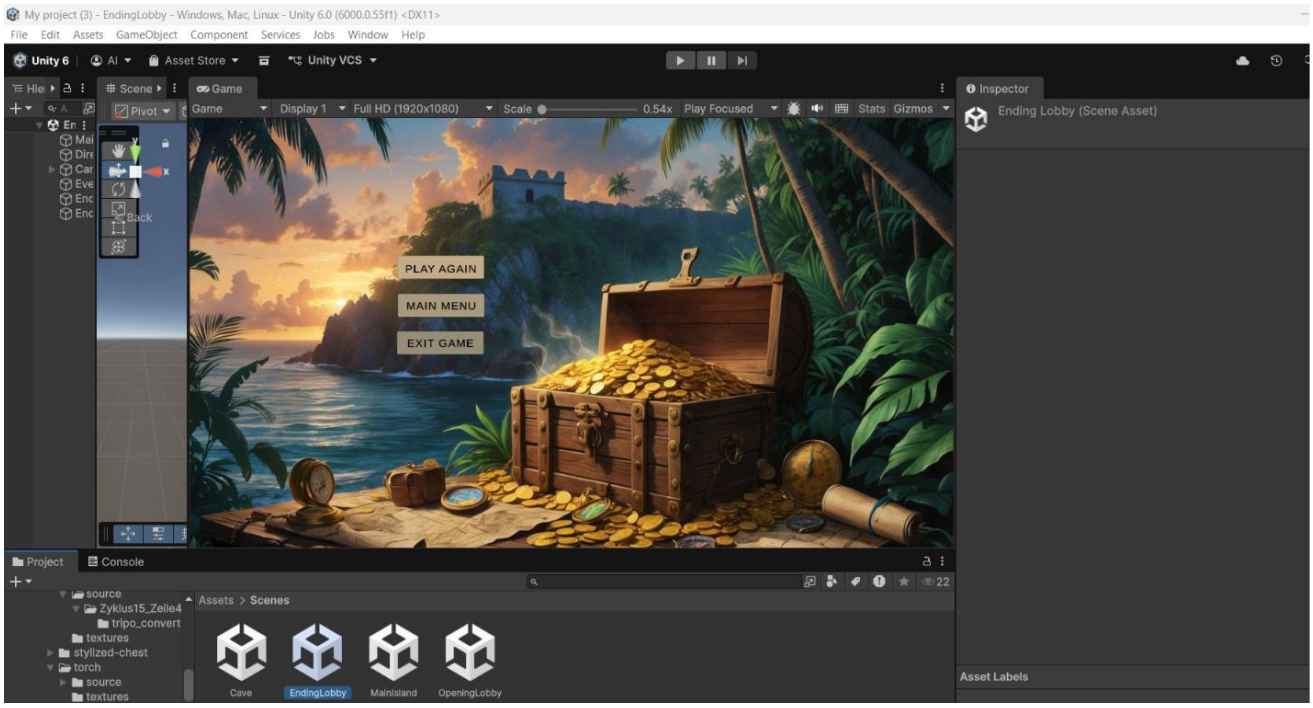
4.1 Player and environment collision

The project uses several Unity systems that need to work together. Environment objects require appropriate colliders; visual geometry alone does not prevent the player or a physics object from passing through it. The beach brick uses Rigidbody physics and a Box Collider, while the main player uses a CharacterController. This distinction supports different interaction needs: physics-based movement for the brick and controlled character movement for exploration.

Input is used in more than one way. The beach brick responds to WASD through applied Rigidbody force, the jungle brick responds to mouse input, and the player character uses keyboard movement. UI buttons trigger animation states and scene navigation. The chest slider sends a continuous value to the chest lid control, while the E key is used to open the interaction interface after the player enters the trigger area.

Scene organisation is important to the integrated experience. The project includes OpeningLobby, MainIsland, Cave and EndingLobby. The lobby buttons call public methods on OpeningLobbyButtons, which load the relevant scenes by name. The Build Profiles scene list must include the scenes that are loaded at runtime. The group work therefore combines individual scene construction, reusable scripts, imported assets, physics components, UI events and scene-loading configuration.





5. Testing, Challenges and Solutions

Testing was carried out at the level of individual modules and during integration. Student 1 tested the brick falling onto the sand, WASD movement at different speed values, collision with a rock and repeat playback of the minifigure walking animation. Student 2 tested the jungle layout, mouse-controlled brick movement and the Stand, Walk and Flip buttons. Student 5 tested the proximity prompt, slider-driven chest movement, nested collider detection, animation control and cursor behaviour. Student 3/4 tested character movement, environment collision, camera following and lobby navigation.

Reported challenges included a physics brick that moved too slowly on the sand, a minifigure that moved out of the camera view during animation, imported models with scale or positioning issues, character collision with imported environment pieces, camera framing, trigger detection across nested colliders, slider interaction conflicting with mouse-look, and scene-loading configuration.

The solutions described in the individual reports included increasing the physics brick speed, using a parent anchor for the minifigure, adjusting imported object transforms and colliders, using a CharacterController for the player, tuning the camera follow offset and smoothing, forwarding child-collider trigger events to a parent controller, separating cursor states during UI interaction, and adding the required scenes to Build Profiles.

7. Reflection

The project provided experience in dividing a larger interactive environment into manageable modules and combining those modules into a connected Unity experience. The team used different approaches to interaction, including Rigidbody force, mouse input, character movement, Animator states, UI buttons, trigger volumes and a slider-driven object transformation. Bringing these systems together highlighted the importance of consistent object scale, collider configuration, input settings, scene naming and testing.

The work also demonstrated that a feature can function in an isolated scene but still require adjustment before it works in the integrated project. Clear handoff procedures, consistent naming, shared testing and version-control practices are important when several people contribute to one Unity project.

Possible future improvements include adding more environmental audio, polishing transitions between the island zones, adding further feedback to interactive objects, improving the treasure reveal with particle effects or lighting effects, adding collectible items or an inventory system, and refining camera transitions. These are proposed enhancements rather than claims about features already completed.

8. AI Code Assistance Statement

The submitted individual reports state that ChatGPT was used as a learning and troubleshooting aid. The reported uses include understanding Unity settings and workflows, troubleshooting input and animation behaviour, learning how to connect UI controls to Animator states, reviewing example C# code, resolving Unity component configuration problems and planning report structure. Student 5's report also identifies Antigravity as an AI-assisted reference tool.

The individual reports describe the students carrying out implementation, adjustments and testing in Unity. AI assistance should be acknowledged accurately and transparently in accordance with the coursework brief. The group should attach the relevant AI conversation transcripts or other required supporting material and ensure that any code or content assisted by AI is clearly identified as required by the module guidance.

9. Conclusion

The Lost Island combines a beach starting area, jungle exploration, character movement, interactive LEGO bricks, ancient environment elements, scene navigation and a treasure-room interaction. The team divided the work into distinct responsibilities and developed the individual systems before bringing them together. The project demonstrates practical use of Unity environment tools, physics, colliders, character control, animation, UI interaction, lighting and scene management.

The final quality of the submission should be supported by screenshots and a final playthrough of the integrated build. Any feature that is not present or does not work in the final integrated version should be described as a prototype, contribution or planned improvement rather than as a completed gameplay feature.

10. Appendix

Figure A1. Final integrated game overview showing the beach and island environment.

Figure A2. Beach starting area with sand, palm trees, rocks and PlayerStart.

Figure A3. Rigidbody LEGO brick Inspector and WASD test on the beach.

Figure A4. Jungle path and vegetation in the integrated project or the Student 2 prototype scene.

Figure A5. Mouse-controlled LEGO brick and its Rigidbody, Box Collider and MouseInput component.

Figure A6. Player Inspector showing CharacterController and PlayerMovement.

Figure A7. Main Camera Inspector showing CameraFollow and the assigned Player target.

Figure A8. Ancient temple or ruins area and relevant trigger/collider setup.

Figure A9. Treasure chest hierarchy and proximity trigger.

Figure A10. Treasure prompt and slider in Game view, plus chest open state.

Figure A11. OpeningLobby and EndingLobby buttons.

Figure A12. Unity Build Profiles showing the scenes included in the build.

Test	Result / evidence
OpeningLobby New Game loads MainIsland	Test in final build
Player moves and remains grounded on the intended route	Test in final build
Player collides with terrain and solid environment objects	Test in final build
Camera follows the player correctly	Test in final build
Beach physics brick responds to WASD and collides with obstacles	Test in final build
Jungle mouse-controlled brick responds to mouse input	Test in final build
Animation buttons trigger the intended states	Test in final build
Treasure trigger displays the prompt and E opens the slider	Test in final build
Slider controls the chest lid without unwanted player/camera movement	Test in final build
EndingLobby Play Again and Main Menu buttons work	Test in final build